

Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{-2} = \frac{dz}{3x^2 \sin(y+2x)} \quad \text{--- (2)}$$

Taking the first two term we get,

$$dx = \frac{dy}{-2}$$

$$\Rightarrow x = -\frac{1}{2}y + c$$

$$\therefore 2x + y = c_1 \quad \text{--- (3)}$$

Now taking first and 3rd term and using (3)

$$\frac{dx}{1} = \frac{dz}{3x^2 \sin c_1}$$

$$\Rightarrow 3x^2 \sin c_1 dx = dz$$

$$\Rightarrow \sin c_1 x^3 = z + c_2$$

$$\Rightarrow x^3 \sin c_1 - z = c_2$$

$$\Rightarrow x^3 \sin(y+2x) - z = c_2$$

The general solution is

$$\Phi[(2x+y), x^3 \sin(y+2x) - z] = 0.$$

Rule III:

Problem 1:

$$\text{Solve } (z^2 - 2yz - y^2)p + (xy + xz)q = xy - xz$$

Solution:

$$\text{Given } (z^2 - 2yz - y^2)p + (xy + xz)q = xy - xz \quad \text{--- (1)}$$

Lagrange's auxiliary eqns are

$$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + xz} = \frac{dz}{xy - xz}$$

$$\Rightarrow \frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{x(y+z)} = \frac{dz}{x(y-z)}$$

$$\therefore \frac{x dx}{z^2 - 2yz - y^2} - \frac{dy}{y+z} = \frac{dz}{y-z} \quad \text{--- (2)}$$

Choosing $1, y, z$ as multipliers we get,

$$\frac{x dx + y dy + z dz}{z^2 - 2yz - y^2 + y^2 + yz + yz - z^2} = \frac{x dx + y dy + z dz}{0}$$

so that $x dx + y dy + z dz = 0$

Integrating we get,

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{C}{2}$$

$$\therefore x^2 + y^2 + z^2 = C_1$$

✶ Taking 2nd and 3rd eqn of (2), we get,

$$\frac{dy}{y+z} = \frac{dz}{y-z}$$

$$\Rightarrow \frac{dz}{dy} = \frac{y-z}{y+z}$$

$$\Rightarrow \frac{dz+dy}{dz-dy} = \frac{y-z+y+z}{y-z-y-z} \quad \checkmark$$

$$\Rightarrow \frac{dz+dy}{dz-dy} = \frac{y}{-z}$$

$$\Rightarrow z dz + z dy = -y dz + y dy$$

$$\Rightarrow y dz + z dy = y dy - z dz$$

$$\Rightarrow d(yz) = y dy - z dz \quad \checkmark$$

$$\Rightarrow yz = \frac{y^2}{2} - \frac{z^2}{2} + \frac{C}{2}$$

$$\Rightarrow 2yz = y^2 - z^2 + C_2$$

$$\therefore 2yz - y^2 + z^2 = C_2$$

Hence the general solution of (1) is

$$x^2 + y^2 + z^2 = \Phi(2yz - y^2 + z^2).$$

Problem 2:

Solve $z(xp - yq) = y^2 - x^2$

Solution:

Given $zxp - z yq = y^2 - x^2$ ——— (1)

Lagrange's auxiliary eqns are,

$$\frac{dx}{zx} = \frac{dy}{-yz} = \frac{dz}{y^2 - x^2} \quad \text{————— (2)}$$

Taking the first two fractions and simplifying we get

$$\frac{dx}{x} = -\frac{dy}{y}$$

$$\Rightarrow \log x = -\log y + \log c_1$$

$$\Rightarrow \log(xy) = \log c_1$$

$$\therefore xy = c_1$$

Choosing x, y, z as multipliers, we get

$$\frac{xdx + ydy + zdz}{zx^2 - y^2z + zy^2 - zx^2} = \frac{xdx + ydy + zdz}{0}$$

so that $xdx + ydy + zdz = 0$

$$\text{or } \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{c}{2}$$

$$\therefore x^2 + y^2 + z^2 = c_2$$

So, the required G.S. is $\phi(xy, x^2 + y^2 + z^2) = 0$

Problem 3:

Solve $(x-y)p + (x+y)q = 2xz$

Solution:

Given $(x-y)p + (x+y)q = 2xz$ ——— (1)

Here Lagrange's auxiliary eqns are

$$\frac{dx}{x-y} = \frac{dy}{x+y} = \frac{dz}{2xz} \quad \text{--- (2)}$$

Taking first two fractions of (2)

$$\frac{dx}{x-y} = \frac{dy}{x+y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x+y}{x-y}$$

$$\Rightarrow \frac{dy+dx}{dy-dx} = \frac{x+y+x-y}{x+y-x+y}$$

$$\Rightarrow \frac{dy+dx}{dy-dx} = \frac{x}{y}$$

$$\Rightarrow ydy + ydx = xdy - xdx$$

$$\Rightarrow -xdy + ydx = -xdx - ydy$$

$$\Rightarrow \frac{xdy - ydx}{x^2 + y^2} = \frac{xdx + ydy}{x^2 + y^2}$$

$$\Rightarrow d\left(\tan^{-1} \frac{y}{x}\right) = d\left(\frac{1}{2} \log(x^2 + y^2)\right)$$

$$\Rightarrow \tan^{-1} \frac{y}{x} = \frac{1}{2} \log(x^2 + y^2) + \log C$$

$$\Rightarrow 2 \tan^{-1} \frac{y}{x} = \log(x^2 + y^2) + 2 \log C$$

$$\Rightarrow 2 \tan^{-1} \frac{y}{x} = \log(x^2 + y^2) + \log C^2$$

$$\Rightarrow 2 \tan^{-1} \frac{y}{x} = \log C^2 (x^2 + y^2)$$

$$\Rightarrow e^{2 \tan^{-1} \frac{y}{x}} = C^2 (x^2 + y^2)$$

$$\Rightarrow \frac{1}{C^2} = (x^2 + y^2) e^{-2 \tan^{-1} \frac{y}{x}}$$

$$\therefore C_1 = (x^2 + y^2) e^{-2 \tan^{-1} \frac{y}{x}}$$

Choosing $1, 1, -\frac{1}{z}$ as multipliers each fraction of (2) is equal to

$$\frac{dx + dy - \frac{dz}{z}}{x - y + x - y - 2x} = \frac{dx + dy - \frac{dz}{z}}{0}$$

so that $dx + dy - \frac{dz}{z} = 0$

$$\Rightarrow x + y - \log z = c_2$$

Hence the general solution is

$$\phi \left[(x^2 + y^2) e^{-2 \tan^{-1} \frac{y}{x}}, x + y - \log z \right] = 0.$$

* Problem 4:

Solve $(x + 2z)p + (4zx - y)v = 2x^2 + y$

Solution:

Given $(x + 2z)p + (4zx - y)v = 2x^2 + y$ — (1)

Lagrange's auxiliary eqns are,

$$\frac{dx}{x + 2z} = \frac{dy}{4zx - y} = \frac{dz}{2x^2 + y} \quad \text{--- (2)}$$

Choosing $-2x, 1, 1$ as multipliers, for each fraction of (2)

$$\frac{-2x dx + dy + dz}{-2x^2 - 4zx + 4zx - y + 2x^2 + y} = \frac{-2x dx + dy + dz}{0}$$

so that $-2x dx + dy + dz = 0$

$$\Rightarrow -x^2 + y + z = c_1$$

$$\therefore x^2 - y - z = c_1$$

Choosing $y, x, -2z$ as multipliers, each fraction

of ②

$$\frac{ydx + xdy - 2zdz}{xy + 2yz + 4zx^2 - xy - 4zx^2 - 2zy} = \frac{ydx + xdy - 2zdz}{0}$$

so that $ydx + xdy - 2zdz = 0$

$$\Rightarrow d(xy) - 2zdz = 0$$

$$\Rightarrow xy - z^2 = c_2$$

So the required general solution is

$$\phi[x^2 - y - z, xy - z^2] = 0$$

Problem 5:

$$x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$$

Solution:

Given $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$ — ①

The Lagrange's auxiliary equations are,

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{-y(x^2 + z)} = \frac{dz}{z(x^2 - y^2)} \quad \text{--- ②}$$

Choosing $x, y, -1$ as multipliers, each fraction of ②, we get,

$$\frac{xdx + ydy - dz}{x^2y^2 + x^2z - y^2x^2 - y^2z - zx^2 + y^2z} = \frac{xdx + ydy - dz}{0}$$

so that $xdx + ydy - dz = 0$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} - z = c_1$$

$$\therefore x^2 + y^2 - 2z = c_1$$

choosing multipliers $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$, we get

$$\frac{\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}}{y^2 + z - x^2 - z + x^2 - y^2} = \frac{\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}}{0}$$

so that $\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0$

$$\Rightarrow \log x + \log y + \log z = \log c_2$$

$$\therefore xyz = c_2$$

The required general solution is $\Phi [xyz, x^2 + y^2 - 2z] = 0$.

*Problem 6:

$$x^2 p + y^2 q = nxy$$

Solution:

Given $x^2 p + y^2 q = nxy$ ——— (1)

Lagrange's auxiliary eqns are

$$\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{nxy} \text{ ——— (2)}$$

Taking first two fractions we get,

$$\frac{dx}{x^2} = \frac{dy}{y^2}$$

$$\Rightarrow -\frac{1}{x} = -\frac{1}{y} + c$$

$$\Rightarrow -\frac{1}{x} + \frac{1}{y} = c$$

$$\therefore \frac{x-y}{xy} = c_1$$

Choosing $-ny, nx, (x-y)$ as multipliers

$$\frac{-nydx + nxdy + (x-y)dz}{-nxy^2 + nxy^2 + nxy^2 - nxy^2} = \frac{-nydx + nxdy + (x-y)dz}{0}$$

so that $-nydx + nxdy + (x-y)dz = 0$

$$\Rightarrow -\frac{ndx}{x} + \frac{ndy}{y} + \frac{x-y}{xy} dz = 0$$

$$\Rightarrow -\frac{ndx}{x} + \frac{ndy}{y} + c_1 dz = 0$$

$$\Rightarrow -n \log x + n \log y + c_1 z = c_2$$

$$\Rightarrow n \{ \log x - \log y \} - c_1 z = c_2$$

$$\Rightarrow n \log \left(\frac{x}{y} \right) - c_1 z = c_2$$

$$\Rightarrow \log \left(\frac{x}{y} \right) - \frac{c_1 z}{n} = \frac{c_2}{n}$$

$$\Rightarrow \log \left(\frac{x}{y} \right) - \frac{x-y}{xy} \frac{z}{n} = \frac{c_2}{n}$$

$$\therefore \log \left(\frac{x}{y} \right) + \frac{y-x}{xy} \frac{z}{n} = \frac{c_2}{n}$$

The general solution is $\phi \left[\frac{x-y}{xy}, \log \left(\frac{x}{y} \right) + \frac{y-x}{xy} \frac{z}{n} \right] = 0$.

Problem 7:

$$(y+zx)p - (x+yz)q = x^2 - y^2$$

Solution: Given $(y+zx)p - (x+yz)q = x^2 - y^2$ — (1)

Lagrange's auxiliary eqns are

$$\frac{dx}{y+zx} = \frac{dy}{-(x+yz)} = \frac{dz}{x^2 - y^2} \quad (2)$$

Choosing $y, x, 1$ as multipliers,

$$\frac{ydx + xdy + dz}{y^2 + xyz - x^2 - xyz + x^2 - y^2} = \frac{ydx + xdy + dz}{0}$$

so that $ydx + xdy + dz = 0$

$$\Rightarrow d(xy) + dz = 0$$

$$\Rightarrow xy + z = c_1$$

Now choosing $x, y, -z$ as multipliers,

$$\frac{xdx + ydy - zdz}{xy + zx^2 - xy - y^2z - zx^2 + y^2z} = \frac{xdx + ydy - zdz}{0}$$

$$\text{so that } xdx + ydy - zdz = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} - \frac{z^2}{2} = \frac{c_2}{2}$$

$$\therefore x^2 + y^2 - z^2 = c_2$$

The general solution is $\phi[xy + z, x^2 + y^2 - z^2] = 0$.

Problem 8:

$$(y-z)p + (x-y)q = z-x$$

Solution: Given $(y-z)p + (x-y)q = z-x$ ——— ①

The Lagrange's auxiliary equations are

$$\frac{dx}{y-z} = \frac{dy}{x-y} = \frac{dz}{z-x} \quad \text{————— ②}$$

Choosing 1, 1, 1 as multipliers, each fraction in ②

$$= \frac{dx + dy + dz}{y-z + x-y + z-x} = \frac{dx + dy + dz}{0}$$

$$\text{so that } dx + dy + dz = 0$$

$$\Rightarrow x + y + z = c_1$$

Again choosing x, z, y as multipliers, each

fraction in ②

$$= \frac{x dx + y dy + z dz}{x(y-z) + z(x-y) + y(z-x)} = \frac{x dx + y dy + z dz}{0}$$

so that $x dx + y dy + z dz = 0$

$$\Rightarrow \frac{x^2}{2} + yz = C_1$$

$$\Rightarrow x^2 + 2yz = C_2$$

Hence the general solution is

$$\phi(x+y+z, x^2+2yz) = 0 \quad \text{or} \quad x^2 + 2yz = f(x+y+z).$$

Problem 9c

$$(y^3x - 2x^4)p + (2y^4 - x^3y)q = 9z(x^3 - y^3)$$

Solution:

Given $(y^3x - 2x^4)p + (2y^4 - x^3y)q = 9z(x^3 - y^3)$ ——— ①

Lagrange's auxiliary equations are

$$\frac{dx}{y^3x - 2x^4} = \frac{dy}{2y^4 - x^3y} = \frac{dz}{9z(x^3 - y^3)} \quad \text{————— ②}$$

Choosing $\frac{3}{x}, \frac{3}{y}, \frac{1}{z}$ as multipliers each fraction

$$\begin{aligned} \text{in ②} &= \frac{\frac{3}{x} dx + \frac{3}{y} dy + \frac{1}{z} dz}{\frac{3}{x}(y^3x - 2x^4) + \frac{3}{y}(2y^4 - x^3y) + \frac{1}{z} 9z(x^3 - y^3)} \\ &= \frac{\frac{3}{x} dx + \frac{3}{y} dy + \frac{1}{z} dz}{3(y^3 - 2x^3) + 3(2y^3 - x^3) + 9(x^3 - y^3)} \\ &= \frac{\frac{3}{x} dx + \frac{3}{y} dy + \frac{1}{z} dz}{0} \end{aligned}$$

so that $\frac{3}{x} dx + \frac{3}{y} dy + \frac{1}{z} dz = 0$

$$\Rightarrow 3\log x + 3\log y + \log z = \log c_1$$

$$\Rightarrow \log(x^3 y^3 z) = \log c_1$$

$$\therefore x^3 y^3 z = c_1$$

Again taking 1st and second fraction in (2), we get

$$\frac{dx}{y^3 x - 2x^4} = \frac{dy}{2y^4 - x^3 y}$$

$$\Rightarrow (y^3 x - 2x^4) dy + (x^3 y - 2y^4) dx = 0$$

$$\Rightarrow (3y^3 x - 2y^3 x - 2x^4) dy + (3x^3 y - 2x^3 y - 2y^4) dx = 0$$

$$\Rightarrow 3y^3 x dy + 3x^3 y dx - 2x(x^3 + y^3) dy - 2y(x^3 + y^3) dx = 0$$

$$\Rightarrow x y (3x^2 dx + 3y^2 dy) - 2x(x^3 + y^3) dy - 2y(x^3 + y^3) dx = 0$$

$$\Rightarrow \frac{3x^2 dx + 3y^2 dy}{x^3 + y^3} - \frac{2dy}{y} - \frac{2dx}{x} = 0$$

$$\Rightarrow d(\log(x^3 + y^3)) - \frac{2dy}{y} - \frac{2dx}{x} = 0$$

$$\Rightarrow \log(x^3 + y^3) - 2\log y - 2\log x = \log c_2$$

$$\Rightarrow \log\left(\frac{x^3 + y^3}{x^2 y^2}\right) = \log c_2$$

$$\therefore \frac{x^3 + y^3}{x^2 y^2} = c_2$$

Hence the general solution is

$$\Phi\left(x^3 y^3 z, \frac{x^3 + y^3}{x^2 y^2}\right) = 0 \quad \text{or} \quad x^3 y^3 = x^2 y^2 f(x^3 y^3 z)$$

Rule IV:

Problem 1:

$$xz^2 + yz^2 = xy$$

Solution:

$$\text{Given } xz^2 + yz^2 = xy \quad \text{--- (1)}$$

The Lagrange's auxiliary eqns are

$$\frac{dx}{xz} = \frac{dy}{yz} = \frac{dz}{xy} \quad \text{--- (2)}$$

Taking first two fractions and cancelling z , we get,

$$\frac{dx}{x} = \frac{dy}{y}$$

$$\Rightarrow \log x = \log y + \log c_1$$

$$\Rightarrow \log\left(\frac{x}{y}\right) = \log c_1$$

$$\therefore \frac{x}{y} = c_1$$

Taking $\frac{1}{x}, \frac{1}{y}, 0$ as multipliers, each fraction of

$$\textcircled{2} \quad = \frac{\frac{dx}{x} + \frac{dy}{y}}{z + z} = \frac{ydx + xdy}{2zxy} \quad \text{--- (3)}$$

Combining the ^{3rd} fraction of (2) and (3), we get

$$\frac{ydx + xdy}{2zxy} = \frac{dz}{xy}$$

$$\Rightarrow d(xy) = 2z dz$$

$$\Rightarrow xy = z^2 + c_2$$

$$\therefore xy - z^2 = c_2$$

$\therefore$ The required general solution is $\phi\left[\frac{x}{y}, xy - z^2\right] = 0$

$$\begin{aligned} \textcircled{2} \Rightarrow \frac{ydx}{2xz} &= \frac{xdy}{2yz} = \frac{-2zdz}{2xy} \\ \Rightarrow \frac{ydx + xdy - 2zdz}{2xyz} &= \frac{ydx + xdy - 2zdz}{2xyz} = 0 \\ \Rightarrow ydx + xdy - 2zdz &= 0 \\ \therefore xy - z^2 &= c_2 \end{aligned}$$

Problem 2:

$$(x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 2(x^2 + y^2)z$$

Solution:

$$\text{Given } (x^3 + 3xy^2)dx + (y^3 + 3x^2y)dy = 2(x^2 + y^2)z \quad \text{--- (1)}$$

Lagrange's auxiliary eqns are,

$$\frac{dx}{x^3 + 3xy^2} = \frac{dy}{y^3 + 3x^2y} = \frac{dz}{2(x^2 + y^2)z} \quad \text{--- (2)}$$

Choosing $\frac{1}{x}$, $\frac{1}{y}$, $-\frac{2}{z}$ as multipliers,

$$\frac{\frac{1}{x}dx + \frac{1}{y}dy - \frac{2}{z}dz}{x^2 + 3y^2 + y^2 + 3x^2 - 4x^2 - 4y^2} = \frac{\frac{dx}{x} + \frac{dy}{y} - \frac{2dz}{z}}{0}$$

$$\text{so that } \frac{dx}{x} + \frac{dy}{y} - \frac{2dz}{z} = 0$$

$$\Rightarrow \log x + \log y - 2 \log z = \log c_1$$

$$\Rightarrow \log(xy) - \log z^2 = \log c_1$$

$$\Rightarrow \log \frac{xy}{z^2} = \log c_1$$

$$\therefore \frac{xy}{z^2} = c_1$$

Choosing 1, 1, 0 as multipliers we get,

$$\frac{dx + dy}{x^3 + 3xy^2 + y^3 + 3x^2y} = \frac{dx + dy}{(x + y)^3} \quad \text{--- (3)}$$

Choosing 1, -1, 0 as multipliers we get,

$$\frac{dx - dy}{x^3 + 3xy^2 - y^3 - 3x^2y} = \frac{dx - dy}{(x - y)^3} \quad \text{--- (4)}$$

From (3) and (4) we get,

$$\frac{dx+dy}{(x+y)^3} = \frac{dx-dy}{(x-y)^3}$$

$$\Rightarrow (x+y)^{-3}(dx+dy) = (x-y)^{-3}(dx-dy)$$

$$\Rightarrow -\frac{(x+y)^{-2}}{2} = \frac{-(x-y)^{-2}}{2} + \frac{C_2}{2}$$

$$\Rightarrow -\frac{1}{2}(x+y)^{-2} + \frac{1}{2}(x-y)^{-2} = \frac{C_2}{2}$$

$$\therefore (x-y)^{-2} - (x+y)^{-2} = C_2$$

The required general solution is

$$\Phi\left[\frac{xy}{z^2}, (x-y)^{-2} - (x+y)^{-2}\right] = 0$$

class

Problem 3:

If u is a function of x, y and z which satisfies the partial differential equation

$$(y-z)\frac{\partial u}{\partial x} + (z-x)\frac{\partial u}{\partial y} + (x-y)\frac{\partial u}{\partial z} = 0$$

Show that u contains x, y and z only in combinations of $x+y+z$ and $x^2+y^2+z^2$.

Solution:

$$\text{Given } (y-z)\frac{\partial u}{\partial x} + (z-x)\frac{\partial u}{\partial y} + (x-y)\frac{\partial u}{\partial z} = 0$$

In this case the auxiliary eqns are

$$\frac{dx}{y-z} = \frac{dy}{z-x} = \frac{dz}{x-y} = \frac{du}{0}$$

and they are equivalent to the three

relations

$$du = 0$$

$$\therefore u = C_1 \quad \text{--- (2)}$$

$$dx + dy + dz = 0$$

$$\Rightarrow x + y + z = c_2 \quad \text{--- (3)}$$

$$x dx + y dy + z dz = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{c_3}{2}$$

$$\therefore x^2 + y^2 + z^2 = c_3 \quad \text{--- (4)}$$

(2), (3) and (4) are the integrals.

Hence the general solution is of the form

$$u = f(x + y + z, x^2 + y^2 + z^2).$$

Problem 4:

$$(y^2 + z^2 - x^2)p - 2xyq + 2zx = 0$$

Solution:

$$\text{Given } (y^2 + z^2 - x^2)p - 2xyq + 2zx = 0 \quad \text{--- (1)}$$

The auxiliary equations are

$$\frac{dx}{y^2 + z^2 - x^2} = \frac{dy}{-2xy} = \frac{dz}{-2zx} \quad \text{--- (2)}$$

Choosing x, y, z as multipliers, each fraction in (1)

$$= \frac{xdx + ydy + zdz}{x(y^2 + z^2 - x^2) + y(-2xy) + z(-2zx)}$$

$$= \frac{xdx + ydy + zdz}{xy^2 + xz^2 - x^3 - 2xy^2 - 2z^2x}$$

$$= \frac{xdx + ydy + zdz}{-x^3 - xy^2 - z^2x}$$

$$= \frac{xdx + ydy + zdz}{-x(x^2 + y^2 + z^2)} \quad \text{--- (3)}$$

Combining the second fraction in ② with fraction

③ we get

$$\frac{x dx + y dy + z dz}{-x(x^2 + y^2 + z^2)} = \frac{dy}{-2xy}$$

$$\Rightarrow \frac{2(x dx + y dy + z dz)}{x^2 + y^2 + z^2} = \frac{dy}{y}$$

$$\Rightarrow \log(x^2 + y^2 + z^2) = \log y + \log c_1$$

$$\Rightarrow \log\left(\frac{x^2 + y^2 + z^2}{y}\right) = \log c_1$$

$$\therefore c_1 = \frac{x^2 + y^2 + z^2}{y}$$

Again taking the 2nd and 3rd fraction in ② we get

$$\frac{dy}{-2xy} = \frac{dz}{-2zx}$$

$$\Rightarrow \frac{dy}{y} = \frac{dz}{z}$$

$$\Rightarrow \log y = \log z + \log c_2$$

$$\Rightarrow \log(y/z) = \log c_2$$

$$\therefore \frac{y}{z} = c_2$$

Hence the general solution of the given equation

$$\text{is } \varphi\left(\frac{x^2 + y^2 + z^2}{y}, \frac{y}{z}\right) = 0$$

$$\text{or, } x^2 + y^2 + z^2 = y f\left(\frac{y}{z}\right).$$

Simultaneous equation of the form $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

The general type of a set of simultaneous equation of the first order having three variables is

$$\left. \begin{aligned} P_1 dx + Q_1 dy + R_1 dz &= 0 \\ P_2 dx + Q_2 dy + R_2 dz &= 0 \end{aligned} \right\} \text{--- ①}$$

where the coefficients $P_1, Q_1, R_1, P_2, Q_2, R_2$ are all functions of x, y, z .

Solving these equations simultaneously, we have

$$\frac{dx}{Q_1 R_2 - Q_2 R_1} = \frac{dy}{P_2 R_1 - R_2 P_1} = \frac{dz}{P_1 Q_2 - P_2 Q_1}$$

which is of the form,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \text{--- ②}$$

where P, Q, R are functions of x, y, z .

Thus, we note that the simultaneous equation ① can always be put in the form ②.

The nature of solution of $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

The given equations are said to be completely solved when we get a solution of the form

$$u_1(x, y, z) = c_1 \text{ and } u_2(x, y, z) = c_2.$$

where u_1 and u_2 are independent integrals (solutions) of the given equation.

u_1 and u_2 are said to be independent integrals if u_1 or u_2 is not merely a constant.

For example, $u_1 = x^2 + y^2 + z^2$ and $u_2 = x + y + z$ are independent whereas $u_1 = zx + zy + zz$ and $u_2 = k(x + y + z)$ are not independent.

Geometrical interpretation of $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

From three dimensional co-ordinate geometry (i.e. solid geometry) it is known that the direction cosines of the tangent to a curve are proportional to dx, dy, dz .

Hence the above differential equation represents curves, the direction cosines of the tangent to the curve at (x, y, z) being proportional to P, Q, R .

Suppose that the solution of the given eqn is given by $u_1(x, y, z) = c_1$ and $u_2(x, y, z) = c_2$.

Then we observe that the solution represents the curves of intersection of the surfaces $u_1(x, y, z) = c_1$ and $u_2(x, y, z) = c_2$.

Since c_1 and c_2 can take any value in infinite number of ways, we get infinite number of such curves.